Carbon Credits Explained—for Engineers and Developers

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Prabhat Vilas Karlekar

Below I have images of research and learning, hand-drawn on A4 paper as a visual guide to understand the carbon credits.

Carbon Credit / Carbon offset

Carbon Credit is unit corresponding to removal of; from atmosphere

1 tonne CO_2 = 1 carbon credit
(1000 kg CO_2) (Certificate/pollution currency)

by Planting trees, or prevent carbon from being emitted.

why? Too much CO2 = earth gets hotter, major contributor of global warming.

- And because many Countries have signed "Paris Agreement", aims,
  - Reduce global warming to under 1.5°C
  - Net-zero emission by 2050
    → ideal is ~0.99°C (preindustry era 1850-1900)

As of now, every yr. ~40+billion tonnes is emitted

Example

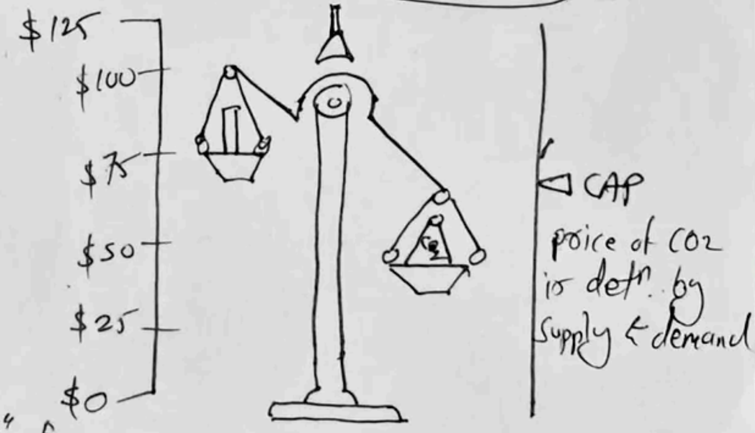
Company A (steel factory)
• Emission Allowed: 100t ~100cc
but emits, 150t
needs 50 more credits, buys

Company B (Solar energy Company)
• Allowed: 100 credits ~100t
• Emits: 20t, saves 80
→ Earns 80 cc

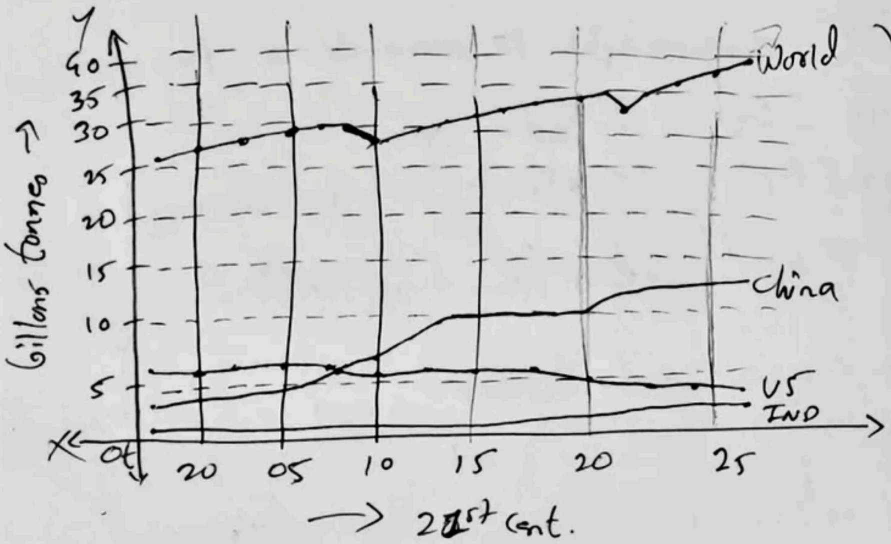
Carbon-Markets

1 → Compliance Market (Gov. Regulated)
  → EU ETS, California C&T, China ETS
  (60$ - $100 per cc)

2 → Voluntary Markets ($1-$15/cc)
  (Anyone can buy)
  → Verra, Gold standard, climate action Reserved
  → Airlines offering "offset your flight" option



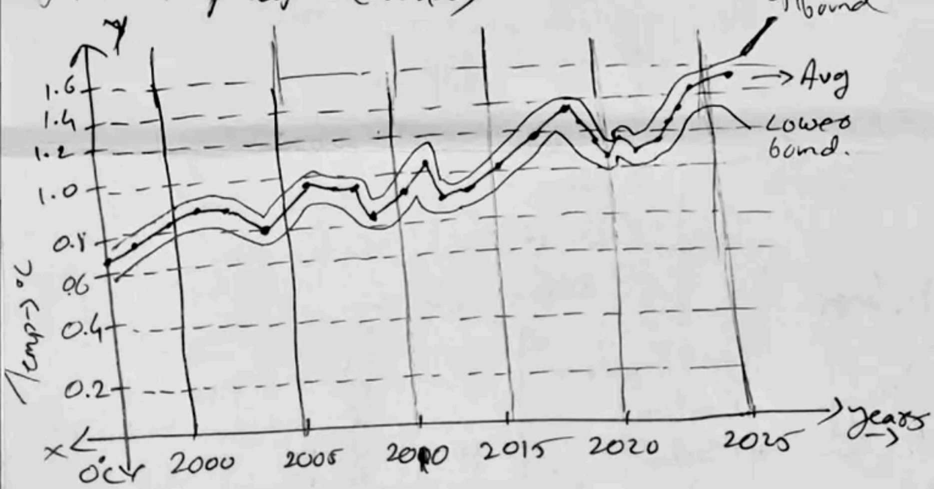
Annual CO_2 emission →



At 2000,
World → 25.5 Bt
US → 6.02 Bt
China → 3.64 Bt
Russia → 1.48 Bt
India → 987.07 mt
Japan → 1.26 Bt
UK → 569.03 mt

At 2024,
World → 38.60 Bt
US → 4.90 Bt
China → 12.29 Bt
Russia → 1.78 Bt
India → 3.19 Bt
Japan → 961.87 mt
UK → 312.91 mt

Annual Temperature (world)



→ India is world's third largest carbon emitter.

Q) How Carbon credit are generated? and economy is created?

(1)

1 Carbon credit = 1 tonne of CO_2, either:

-> Avoided (never release into atmosphere)
or
Removed (pulled out of the atmosphere)

- Digital Certificate,
- like stock share or crypto-currency token

Certificate
ID:

Certificate is
-> Tradable (bought/sold)
-> Unique (has num)
-> Retirable (once used, cannot be reused)
How ?

? Who actually generates carbon credits.

=> Not Verra or GS or VCS, real-world projects on ground does. Examples:-

Project Type	Example	How it create Carbon credits
• Renewable energy	Wind farms, Solar plants etc.	- Reduce/replace Coal/gas power ——> avoids emission
• Efficient Cookstoves	Honduras cookstove projects	- Reduces firewood/charcoal use
• Reforestation/ Afforestation	Madagascar, Brazil planting projects	- Plants trees that observe CO_2
* • Direct Air Capture (DAC)	Climeworks (Switzer-land), Carbon Engineering	- Machine that sucks CO_2 from the air.

★ Can generate credits only if: -> Reduces/remove emission Vs a baseline scenario

-> Only if reduction is:-
-> Additional (
-> Real
-> Measurable, and
-> Verified by independent auditor (VCS) (/VBB)

- It won't happen anyway without Carbon finance
- e.g poor households wouldn't afford stoves without carbon finance.

So, "Projects generate the climate impact", and "Standards (Verra/GS) convert that impact into tradable credits"

Q3 What do Verra/Gold Standard actually do! ACR, CAR

=> (2)

They are:

Standard Bodies/ Rule-makers → 3.1

- Publish Methodologies (rulebook)
  - -> Define base (what would happen without the project)
  - -> How to measure data (fuel use, energy reduction, tree growth etc)
  - -> How to calculate emission and reductions
  - -> How to deal with: leakage (emission shifted elsewhere)
    - Uncertainty
    - Risk buffers (for forest burning etc)

Trusted Referees

Mint Credit printer

Registries (bank + ledger)

3.2 Register/Approved

-> Project developers submits:

- Project design Document (PDD)
- Baseline study, monitoring plan and chosen methodology.

Verra/GS

-> check Eligibity (Review every project document and every calculation)

through: Third-party auditors (VVBs)

3.3

ie Validation & verification Bodies

- -> Validate project design (before operation)
- -> Verify emission reductions periodically (after data is collected)

3.4; Issue credits in their registry,

-> Post monitoring & verification

"Project reduced X tCO_2e in period Y"

Verra/GS issue X credit

Therefore credit became digital assets that can be traded.

Example:

Issuance (Minting)

Case study: Rwanda

Clean Cookstove pgm (Reg. under GS/Verra)

Metric	Value
Household served	12,000
Baseline wood use	3.5 t/HH/yr
improved Stove wood use	1.2 t/HH/yr
Emission factor	1.8 tCO_2/tonne wood
Net wood saved	2.3 tonne/HH/yr
Annual CO_2 reduction	49,680 tCO_2

=> CO_2 saved per household = 2.3×1.8 = 4.14 tCO_2/yr

Total = 4.14 × 12,000 = 49,680 tCO_2/yr $\xrightarrow[\text{tradeable credit}]{\text{become issuable}}$ 49680

Component (3)	Who	What they do	Output (VCM)
- Project Developer	Private Companies, Energy firms & forest owners etc (Avoidance) Ex: EcoSafi (Kenya), India Panna ARR (Removal) Microsoft bought 1.5t	Run real projects that avoid/Remove CO_2 from atmosphere	Actual climate impact (measured in tonnes)
- ~~Verra/Gold Standard~~ Methodology	Verra/Gold Standard	Define rules, regulations and formulas to cal. CO_2 reduction/avoidance	Standardized Calculation methods
- VVB (Auditors)	TUV, KPMG, Earthood, etc	Independently verify, monitoring data	Verified tonne CO_2
Verra/Gold standard	provide standard/ and + Registry	Review, approve and issue carbon credits	Digital Carbon credits/ Certificate (VCUs or GS VERs)
Market Participants	Brokers, traders, Corporates	~~By~~ Buy & sell credits (carbon exchange platform)	Price discovery, liquidity
Buyers	Google, Microsoft, Shell, airlines, individuals	Retire credits to offsets emissions	Climate claim (carbon neutral etc.)

Voluntary Market

(4) Economy Creation:

Element	Descriptions	Real Numbers
Supply	Credits issued by different projects	300 M Credits/year globally.
Demand	Corporate offsets, net-zero targets	> $2.3B annual voluntary market
Price range	Varies by project type	$1 - $1200 / credit
Liquidity channels	Brokers, OTC deals, Carbon neutral Xpansiv, South pole, climate impact partners,	—
Primary ~~sells~~ sale	Developer sells Credits	$6 - $20 typical for Cookstoves
Secondary Market	Traders resell Credits	$20 - $40 (depends on demand)
Retirement	Buyer cancel Credit	Remove it from tradable support / Burn them.

Example:- say sale = $11/credit, Net credit = 49680

Revenue = 49 680 × $11 = $ 5,46,480

Distribution:-

Stakeholder	share
Project developer	65%
Local NGO partner	10%
Monitoring & audits	8%
Verra/GS reg. fees	2-3%
Sales broker	10-12%

Clean cookstove project become financially viable only because Carbon credits created this revenue.

Climate Impact → Tradable Credits

(5) Cap-And-Trade (Compliance Market)

Component	Who	What they do	Output
Regulator / Government	EU, CARB, KOREA MOE, California air resources board	Define annual cap, issues allowances, enforce penalties	EUAs, CCA, KAU (Official emission allowances)
Compliance Entities	Power plants, oil refineries, steel, airlines.	Must surrender allowances equal to their emission	Demand for allowances
Registry operator	EU CITL, Union registry, CARB registry.	Ledger of who owns which allowances	Digital allowance accounts.
MRV Bodies (Monitoring, Reporting, Verification)	Accredited verification auditors	Verify emissions from each factory	Verified emission (tonnes CO_2)
Exchanges	ICE, EEX, KRX	Trading of allowances and futures	Price ~~disto~~ discovery
Brokers & Traders	Bank, funds intermediaries	Liquidity and arbitrage	Volume & liquidity

Why free allocation? (7)
→ Some indust. compete globally
Ex: steel, cement, aluminium, fertilizer
If EU forces them to ~~buy~~ buy 100% of allowances, their cost goes up - they might
→ Move outside EU,
→ Produce in countries with cheaper/no carbon pricing.
This is called carbon leakage

So EU gives free allowances to indust. at risk of moving production outside.
cause, if it moves then global emission increase, not decrease.

How Cap is decided for industries?
→ EU analyse 2 major factors:-

Factor	Meaning
Trade intensity	How much industry competes with the import/export globally
Carbon cost intensity	Carbon price impact as % of their value

If both are high → receives more free EUAs

Ex:- Cement plant in Italy
Produces: 1,00,000 tonnes of clinker
Benchmark: 0.766 t CO_2 per tonne
Leakage factor: 100% (cement = high-risk)
So free EUAs:
1,00,000 × 0.766 × 1.00
→ 76,600 EUAs (free)
If plant emit → 90,000
→ must buy 13,400 EUAs
If 1 EUA = €80 × 13400
then ~~1.72€~~ € 1.72 million annual cost
This pushes them to reduce emission, but not enough to leave from EU

Example:- (6)
(1) Govt./Reg. set the CAP
→ EU set limit on total emission
Say FY24 → 1.5B t → 25 → 1.1B
(2) Regulators "Mints" Allowances
→ Creates 1.5B EUA in Union reg.
Each EUA = permission to emit 1 tonne CO_2
(3) Distribution:
→ free allocation to industries at risk
→ Auctions via EEX
(4) Companies Emit CO_2
→ power plant, refineries measure their emissions
(5) Monitoring / Reporting / Verification (MRV)
→ Confirms actual emission
(6) Compliance Surrender
→ Companies must surrender allowances equal to emission
Emit 10,000 t CO_2 - must submit 10,000 EUAs
If short → buy from market
If excess → sell or bank for next yr.
(7) Trading on Market:
Ex. { ICE (Intercontinental Exch.)
EEX (European Energy Exch.)
(8) If a Company does not surrender enough EUAs.
→ € ~~100~~ 100 penalty per tonne
→ Must still surrender missing allowances next year.

Visual explanations make complex systems easier hope these diagrams helped.

Thank you!

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🔗 Portfolio: https://prabhatkarlekar.github.io/
💻 GitHub: https://github.com/PrabhatKarlekar
🔗 LinkedIn: https://linkedin.com/in/prabhatkarlekar

Carbon Credits | Carbon Emissions | Economy | Software Engineering | Design

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Written by Prabhat Vilas Karlekar

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Aspiring SDE with 3+ years' experience in development and QA, working on ML, cryptography, blockchain, and CI/CD to build secure, scalable systems.

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Before starting, I want to tell you all that this article-paper was not in our planned activities, it came out when we started learning...

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